Example Queries

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Below are examples of supported questions according to the user profile and expertise.

Production Manager (Simulation & Predictive Maintenance)

As a **production manager**, I need to establish a performance baseline, plan a specific target production, and assess the impact of a machine breakdown.

1. *Run 5 simulation replicas starting from an empty initial state for a duration of 1000 seconds to check how many pieces are produced.*
2. *Perform 5 simulation replicas with a warm-up of 200 seconds. After the warm-up, simulate the production of 100 pieces and report if they can be produced in 500 seconds.*
3. *Simulate the production over 3000 seconds starting from an empty initial state with 5 replicas. If `station_51` fails, what is the mean processing time for producing 200 pieces?*

Safety Engineer (Formal Verification)

As a **safety engineer**, I want to audit the production system by understanding the automaton semantics (locations/events), verifying global safety (i.e., deadlocks), and checking specific temporal behaviors.

1. *Return the events and locations in the automaton.*
2. *What does location q_6 mean?*
3. *What does event s_6 mean?*
4. *Is event s_6 reachable?*
5. *Check if location q_6 is reachable within 15 seconds.*
6. *Check if the process contains any deadlocks.*
7. *Check if the process will always eventually reach location q_1.*

Process Analyst (Process Mining)

As a **process analyst**, I have a raw event log and I want to discover the underlying Petri net, check if reality matches the model, and then analyze execution frequencies.

1. *Can you discover the Petri net from the provided event log?*
2. *Is the event log conformant with the behavior prescribed by the Petri net?*
3. *What is the most frequent execution variant recorded in the log?*
4. *What is the workload of `station_31` in the log?*

Plant Manager (Multi-Perspective Questions)

As a **plant manager**, I need to make a real-time decision by combining a safety check with a performance forecast in a single step.

1. *Please check whether there is a deadlock. Then, assuming `station_51` fails, run 10 simulation replicas for 15000 seconds from an empty initial state to check how many pieces can be produced.*